

!->The layer “NOAA Storm Events Database 1950-2025” by dpisut_content appears to only contain data up the end of 2024. The feature name in the service URL is “NOAA_Storm_Events_Database_1950-2021_v2”, the feature alias in ArcGIS Pro is “Storm_Events_2021_Final”, and the AGOL service webpage says the dataset was updated on Jan 15, 2026. The Service used for this assignment is: https://services.arcgis.com/jlL9msH9OI208GCb/arcgis/rest/services/NOAA_Storm_Events_Database_1950-2021_v2/FeatureServer

For Part 2, question 3, I assume we are meant to discuss cluster analysis tools (instead of network analysis tools) used in this paper?<--!

Part 1

Q1: When using the non-detailed Texas state boundary layer (https://services.arcgis.com/KTcxiTD9dsQw4r7Z/arcgis/rest/services/Texas_State_Boundary/FeatureServer) there are 9,723 tornados selected in Texas in the dataset.

Q2: Using the summarize tool (with case field = Year, statistics field = Year, and statistics type = count) I know that 2015 had the highest number of tornadoes at 260.

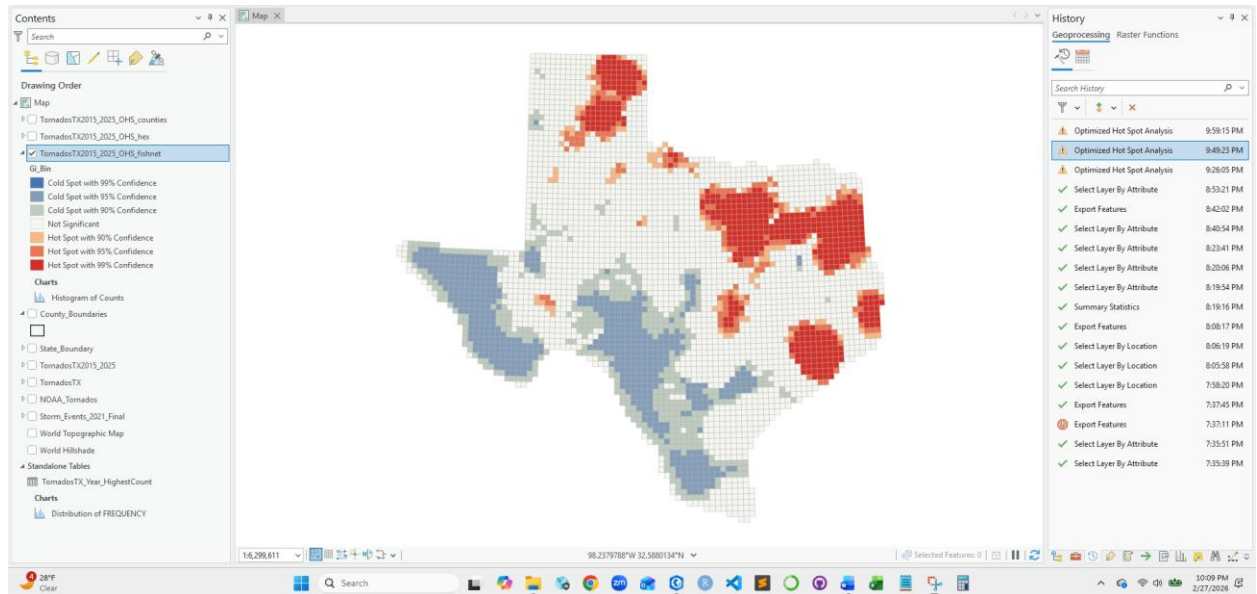
Q3: With the summarize table created for question two, I visualize statistics to understand the frequency distribution of tornados per year, and get a mean value of 129.64 tornadoes (9723/75).

Q4:

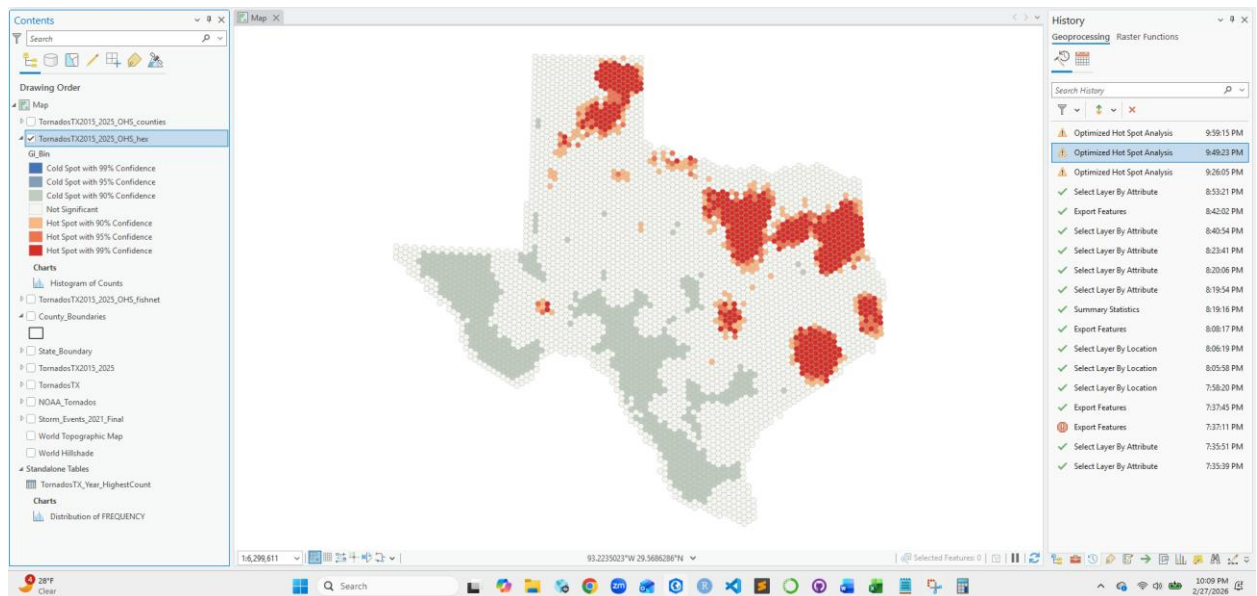
- Fishnet: The optimal fixed distance band is based on the average distance to 30 nearest neighbors → 38,652 meters.
- Hexagon: The optimal fixed distance band is based on the average distance to 30 nearest neighbors → 44,942 meters.
- County Polygons: The optimal fixed distance band is based on peak clustering found at 336,361.4066 meters.

Q5:

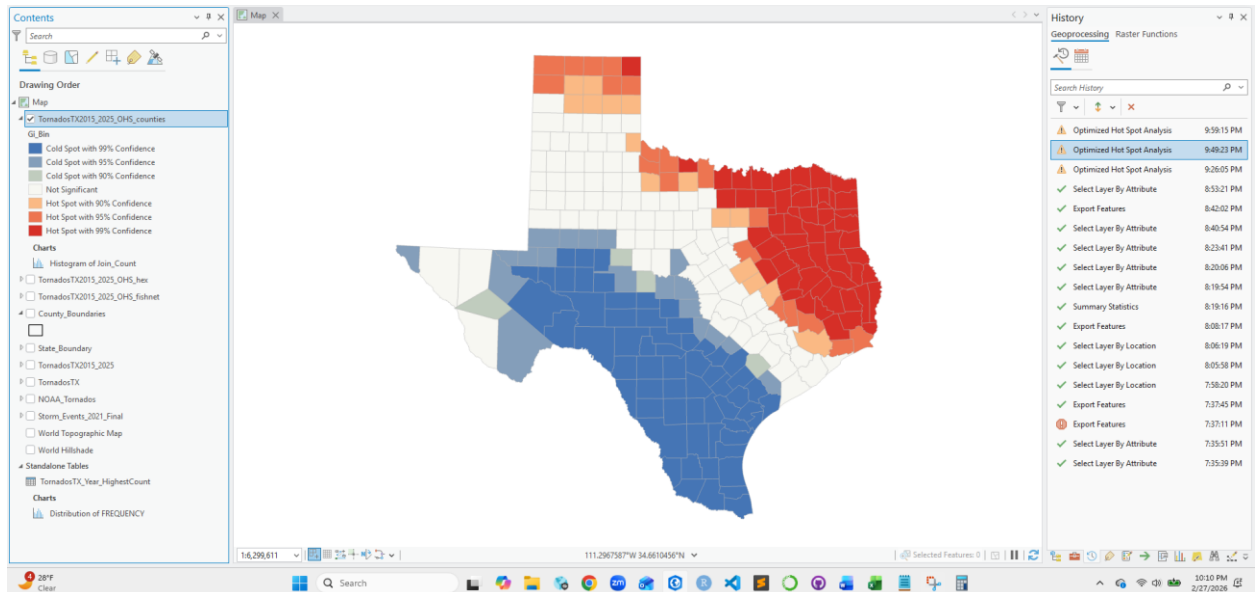
- Fishnet:



- Hexagon:



- County Polygons:



Each of these optimized hot spot analysis outputs show a different distribution visualization of the tornado point data. The fishnet grid has the smallest distance bands and shows a wider range of confidence classes than the hexagon aggregation. The hexagon aggregation has larger distance bands and similar pattern of mapped hot spots with 99% confidence, though with smaller, more precisely defined, clusters overall and without the 95% confidence cold spot areas in the south west region. The county polygon aggregation has the widest range of confidence classes from 99% confidence cold spots to 99% confidence hot spots. In order to best represent curves in the patterns of this data, to reduce sample size bias, and since Texas is a large area with tornado events happening all across it, I think the hexagon aggregation is the best choice to present.

Q6: The hexagon aggregation output shows a clear predominance of hotspots across the eastern and north eastern regions of Texas. There is also a significant hotspot area in the most northern part of Texas, as well a comparatively small, and isolated, hotspot area in west Texas.

Part 2

For the selected reading portion, I choose *“Unequal residential exposure to air pollution and noise: A geospatial environmental justice analysis for Ghent, Belgium”* because this paper falls in line with the kind of research I would like to pursue throughout my MsGIS program at Clark. I am interested in reading about different applications of GIS methods used at municipal and state level community planning to help improve quality of life for residents and advance sustainable development.

Spatial auto-regression models and the use of the Bivariate Moran’s I statistic to calculate coefficients for air and noise pollution exposure to determine if there is a spatial correlation between

them were used (in addition to the ordinary least squares regression models) to control for the effects of spatial autocorrelation (Tobler's first law of geography) on the results of the analysis. Without accounting for spatial autocorrelation, the model would incorrectly demonstrate stronger relationships between predictors that may not actually exist. Spatial interpolation (RIO-interpolation technique) was also used to create the air pollution maps, along with IFDM model to visualize local variations in air quality data caused by traffic. Together, the RIO-IFDM creates a network of NO₂ emission areas based on roadways (not taking industry or household emissions into account, or controlling for "canyon effect"). Other cluster analysis methods included bilinear interpolation to calculate a more accurate estimate of residential exposure by controlling for local differences of household density and Local indicators of Spatial Association (LISA) which is applied to detect spatial patterns such as clusters of high and low values and spatial outliers.

Figure 2 on page 4 is a map displaying the annual mean NO₂ concentration throughout Ghent in 2013. I like how the map is floating in white space because it makes the study area stand out without needless/distracting background noise. The color ramp used is also clear and intuitively understood with dark red being the heaviest concentrations. I think the GIS analyst also made good use of the white space in this map, with correct layout orientation and effective use of visual hierarchy. Although the minimalist stylistic approach is good for highlighting the study area, I think including the roadways as a line feature on this map would improve the reader's ease of understanding the relationship between primary roadways and air pollution concentration. For instance, the map included in figure 1 differentiates roadway type and when I flip the pages back and forth to see both maps, I get a better sense (visually) of how the primary roadways are spatially related to the heaviest concentrations of air pollution as compared to the secondary and tertiary roadways. Also, considering how roadways are essential to the analysis that produced this map (RIO-IFDM model), I think it is appropriate to include that layer as well for orientation.

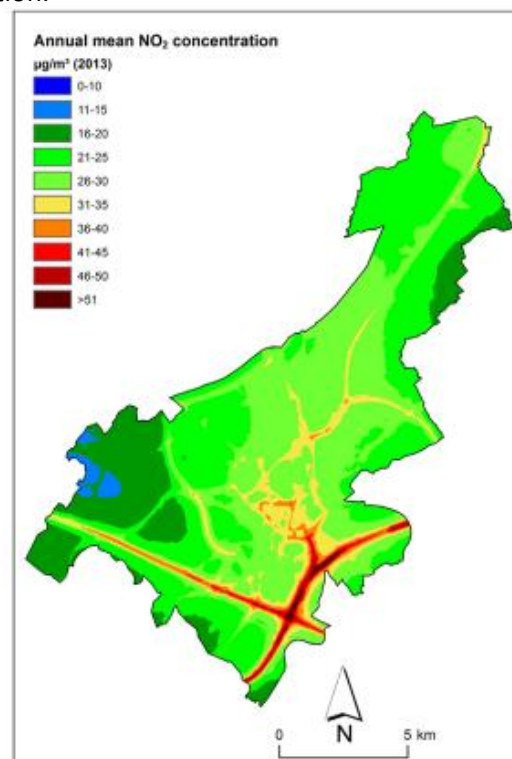


Fig. 2. Distribution of annual mean NO₂ concentration (2013) (based on: ATMOSYS, 2013).